

Shweta Choudhary

Prime Minister Research Fellow



Personal Details

Date of Birth 13 Nov 1999

Nationality Indian

Address Department of Chemistry,
Indian Institute of Technology Roorkee,
Haridwar-247667, Uttarakhand, India

Mobile +91-7015195622

Email shweta_c@cy.iitr.ac.in

Area of Computational chemistry
research



Education

Dec 2022 **PhD in Computational Chemistry**, *Department of Chemistry, Indian Institute of Technology Roorkee, India*

2020–2022 **Master of Science in Chemistry**, *Department of Chemistry, Indian Institute of Technology Roorkee, India, CGPA: 8.88/10*

2017–2020 **Bachelor of Science with honor in Chemistry**, *Department of Chemistry, Gargi College, University of Delhi, India, CGPA: 9.23/10*

Research Experience

2021–2022 **Graduate student**, *“First-principle investigation for anode material for beyond Li-ion batteries, focusing on designing low-dimensional anodes for Na-ion batteries”*, Indian Institute of Technology Roorkee, India

Advisor: **Prof. T. J. Dhillon**

Publications

2026 Dixit, D., **Choudhary, S.**, Sarkar, S., Banerjee, S. & Bhowmik, T. Experimental and computational investigation of selective removal of ciprofloxacin by HKUST-1. *Journal of Water Process Engineering* **86**, 109802 (2026)[[DOI](#)]

- 2026 Goyal, N., **Choudhary, S.**, Witulski, B., Yan, S., Singh, B., Kornienko, N., Banerjee, S. & Mathur, S. Selective Dinitrogen Electroreduction to Ammonia on Zn₂V₂O₇ Pyrochlore. *ACS Catalysis* (2026)[[DOI](#)]
- 2025 **Choudhary, S.** & Banerjee, S. Ion coordination and migration mechanisms in alkali metal complex borohydride-based solid electrolytes. *Communications Chemistry* **8**, 123 (2025)[[DOI](#)]
- 2025 Saroha, R., **Choudhary, S.**, Brahma, R. & Banerjee, S. Designing superatomic (Li/Na/K)-Al-Sc cluster catalysts for selective electrochemical HER and NRR. *The Journal of Physical Chemistry C* **129**, 6562–6573 (2025)[[DOI](#)]
- 2024 **Choudhary, S.**, Saroha, R. & Banerjee, S. Efficient electron injection into graphullerene enables reversible NaC₂ sodium storage. *ACS Applied Materials & Interfaces* **16**, 50859–50869 (2024)[[DOI](#)]
- 2023 Saini, V.[†], Krishankant[†], **Choudhary, S.**[‡], Gaur, A., Banerjee, S., Bagchi, V. & Venkatesh, V. Atomically precise copper nanoclusters as a potential catalyst for the electrochemical oxygen evolution reaction. *Journal of Materials Chemistry A* **11**, 24754–24763 (2023) [‡]Second author[[DOI](#)]
- 2022 **Choudhary, S.**, Duhan, N. & Dhilip Kumar, T. J. Hydrogen passivated β_{12} -borophene nanoribbon: a propitious one-dimensional metallic anode for sodium-ion rechargeable batteries. *Applied Surface Science* **606**, 154825 (2022) [[DOI](#)]

Workshops, Presentations & Conferences

- Mar 2026 **Global Virtual Symposium: Computational Materials Discovery**, ACS Spring 2026. (Oral talk) [[Certificate](#)]
- Jan 2025 **Evolution of Electronic Structure Theory & Experimental Realization (EESTER)**, SRM Institute of Science and Technology & IIT Madras. (Attended workshop & Presented poster) [[Certificate](#)]
- Dec 2024 **Conference on Advances in Chemistry for Energy and Environment (CACEE)**, TIFR Mumbai. (Presented poster) [[Certificate](#)]
- Nov 2024 **Research Scholar's Day-ChemDay**, Department of Chemistry, IIT Roorkee. (Received best poster award) [[Certificate](#)]
- Oct 2024 **CRSI-ACS Early Career Researchers' Symposium**, KIIT Bhubaneswar. (Received best poster award) [[Certificate](#)]

Jan 2024 **International Conference on Functional Materials**, IIT Kharagpur.
(Presented poster) [[Certificate](#)]

Awards, Fellowships & Academic Achievements

2024 Best Poster Award, ChemDay 2024 at IIT Roorkee
2024 Best Poster Award, CRSI-ACS Early Career Researchers' Symposium 2024 at KIIT Bhubaneswar
2023 Prime Minister Research Fellowship (PMRF)
2022 Qualified Graduate Aptitude Test in Engineering (GATE)
2020 Qualified IIT- Joint Admission Test (JAM)

Experience with Theoretical Models, Methods & Packages

Models & methods Density functional theory (DFT), Harmonic and quasi-Harmonic approximation, density functional perturbation theory (DFPT), nudged elastic band (NEB) method, growing string method, classical and *ab initio* molecular dynamics simulations

DFT-based Packages VASP, FHI-aims, FHI-vibes, i-PI, Gaussian, GaussView, Quantum Espresso, Critic2, Material Studio, Avogadro, CP2K, and GROMACS

GUI and Python modules Pymatgen, ASE, VASPKIT code, baderVis, VTST tools, VESTA, VMD, sumo, pyband, Aimstools, GIMS, Phonopy, aimsChain

Technical skills Anaconda, Python, Bash/Shell scripting, LaTeX

Architectures Linux, Mac, Windows

Referees

Prof. Swastika Banerjee
Assistant Professor
Department of Chemistry
IIT Roorkee
sbanerjee@cy.iitr.ac.in

Prof. T. J. Dhilip Kumar
Associate Professor
Department of Chemistry
IIT Ropar
dhilip@iitrpr.ac.in